F79SP Coursework: Manufacturing Processes

Throughout let D be the last digit of your Heriot-Watt student ID number (i.e. the 8 digit number beginning with an H). D will be a number between 0 and 9. For example, if your student ID is H12345678 then $D=8$.

Background

A factory is keen to assess the profitability of a product which undergoes the following manufacturing process. The manufacturing process forms a Markov chain defined on the state space $S=\{A,B,C,P_e,P_s,P_d\}$. The manufacturing process must start by allocating an input item into either Machine A or B. The item will then undergo transitions following the Markov chain, possibly going through Machine C and will eventually lead to the production of a final product, in one of the three categories: excellent, satisfactory and defective, indicated by P_e, P_s and P_d respectively. The manufacturing process's transition matrix is given below, following the order as given in the listing of the elements of S above.

$$P=\begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0.5 & 0.2 & 0.2 & 0 & 0 & 0.1 \\ 0 & 0 & 0.8 & 0.2(0.6+\frac{D}{50}) & 0.2(0.2-\frac{D}{100}) & 0.2(0.2-\frac{D}{100}) \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The factory has proposed the following three initial distributions for consideration:

- **Proposal 1**: All input items start from Machine A.
- **Proposal 2**: All input items start from Machine B.
- **Proposal 3**: All input items start from either Machine A or Machine B at random, with equal probability.

The cost for each final product consists of the following:

- Material cost of \$2 per input item.
- Manufacturing cost of \$0.1 per unit of time per input item in Machine A and in Machine B, and \$0.2 per unit of time per input item in Machine C.

The selling price is \$9 for an excellent product, \$6 for a satisfactory product and a defective product cannot be sold. The manufacturing time of an input item is defined as the total time for the process to hit either one of the P_i's, for $i\in\{e,s,d\}$. The total manufacturing cost of an input item is the sum of the manufacturing costs over its manufacturing time.

Question

Your task is to write a report which clearly indicates the best proposal, in terms of profitability and the manufacturing time. Your report should contain the following main points:

- The transition diagram of the manufacturing process
- The expected manufacturing time of an input item
- The expected proportion of excellent, satisfactory and defective products produced
- The expected total manufacturing cost of an input item
- The expected profit of a final product

You should submit a separate R script which calculates the required quantities (see below) and produces all plots used in your report. Your written report is worth 24 marks and your R script is worth 16 marks so the total assignment will be worth 40 marks.

R code

You are asked to write R code which allows you to do the necessary calculations to write your report. Your R code should be saved as an R script (a '.R' or '.r' file) and not as a Jupyter Notebook or an RData file.

In particular, you must calculate 16 quantities `Quantity1`,..., `Quantity16` which give the following:

- `Quantity1` is the probability an excellent product is eventually produced, given that the input item is currently in Machine C.
- `Quantity2` is the probability a defective product is eventually produced, given that the input item is currently in Machine C.
- `Quantity3` is the expected manufacturing time of an input item under Proposal 1.
- `Quantity4` is the expected manufacturing time of an input item under Proposal 2.
- `Quantity5` is the expected proportion of excellent products under Proposal 1.
- `Quantity6` is the expected proportion of satisfactory products under Proposal 1.
- `Quantity7` is the expected proportion of defective products under Proposal 1.
- `Quantity8` is the expected proportion of excellent products under Proposal 2.
- `Quantity9` is the expected proportion of satisfactory products under Proposal 2.
- `Quantity10` is the expected proportion of defective products under Proposal 2.
- `Quantity11` is the expected total time spent by an input item in Machine A and B under Proposal 1.
- `Quantity12` is the expected total time spent by an input item in Machine A and B under Proposal 2.
- `Quantity13` is the expected total manufacturing cost of an input item under Proposal 1.
- `Quantity14` is the expected total manufacturing cost of an input item under Proposal 2.
- `Quantity15` is the expected profit of a final product under Proposal 1.
- `Quantity16` is the expected profit of a final product under Proposal 3.

Each quantity is worth 1 mark. These will be assessed by running your R code and comparing the results with the verified correct output. You must ensure that you have defined the variables as above (with exactly the same naming and capitalisation) and ensure your code runs without generating errors. **Failure to do so may result in 0 marks being awarded**. You can check your code runs correctly by running it through the file `TestSubmissionAssignmentF79SP.R`. Please do not give references to directories on your own computer (e.g. specifying a directory with setwd) or make use of any R library

Report Specification

Your report should provide clear recommendation on the best proposal for the manufacturing process, in terms of profitability and the manufacturing time. You can make use of the quantities calculated above and any relevant tables or plots to support your argument. The report should meet the following specifications:

1. The main body of the report must consist of the following four sections: Introduction, Model and Simulation Explanation, Results and Conclusion. They should give the results of the analyses that have been carried out and your comments on and conclusions from these results, but should be as readable as possible by non-specialists.
2. An Appendix section must be included after the main body of the report. The appendix must contain all relevant mathematical details not included in the main report to support your recommendation.
3. The report must be typed.
4. The report must not be more than six sides of A4 including graphs and tables, and also including the Appendix (e.g. if your report is 4 pages, then your appendix should be no more than 2 pages if it starts on a new page).

A total of 24 marks is awarded based on the written report (including the appendix). The following qualities will be taken into account in awarding marks for the written report:

- **Structure and organisation of the report (4 marks):** Your exposition should flow logically without jumps or gaps. You will have succeeded if the reader is always able to understand what you are doing and why, without having to refer forwards in your text.
- **Any tables or figures (4 marks):** In order to assist the reader's understanding, tables and figures should be neat and clear, have an adequate caption, and be clearly referred to in the main text of your report.
- **Validity of your analysis and arguments (7 marks):** Your report should demonstrate clear and logical analysis supported by appropriate simulation results with well-structured arguments. Any statement, especially when relating to a mathematical or statistical quantity should be precise, unambiguous and based on well-defined quantities.
- **Usefulness of your conclusions (2 marks):** The reader is reading your report in order to learn your conclusions, assess the strength of support for them, and in all probability to decide upon some subsequent action that may depend upon them. Can the reader make decisions with confidence based upon what you have written?
- **Technical details (7 marks)**: In the Appendix, you must provide key technical ideas of the mathematical models you have used. You are also required to provide the analytical solutions to the three proposals. Your appendix should cover the following:
  - The expected manufacturing time of an input item
  - The expected proportion of excellent, satisfactory and defective products produced
  - The expected total manufacturing cost of an input item
  - The expected profit of a final product

  These solutions may be used to further strengthen the arguments in your main report. You should avoid including overly detailed calculations, such as demonstrating the method of solving a system of linear equations due to the page limit constraint.

Further Points to Note

- **Questions for the Lecturers**: If you have any questions on the assignment, please ask in the discussion board rather than e-mailing lecturers directly. That way all students will see the same responses.
- **Collaboration and Plagiarism**: Students are allowed to discuss the methods used with other students, but your submitted project must be **all your own work**. Particularly please note the following:
  - Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
  - Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course or worse.
  - Students must never give hard or soft copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
  - Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.

  More information on plagiarism can be found here
- **Submission**: The report and the code should be submitted through Canvas using the appropriate links. It should not be submitted by email or handed in as a paper copy. You should also submit the Student Declaration of Authorship at the same time to the appropriate link.
- **Deadline**: The deadline for submission is 9:00 am UK time = 1:00 pm UAE time = 5:00 pm Malaysia time on Friday 7th November 2025; projects may be submitted early. Late projects will be marked in line with the University Coursework Policy, i.e. a standard deduction of 30% if submitted within 5 working days and 0 marks awarded if submitted later. You can, however, submit a mitigating circumstances application, but these may only be applied at the end of the semester. No individual extensions are allowed.
- **Feedback**: It is expected that feedback on this assignment will be available by Friday 28th November.
- **Assessment**: This coursework will contribute 20% to the final mark for the course.